

Automatic POWER-RESUMPTION ALARM

N. PUGALUM PERUMAL

Most power-resumption alarm devices produce non-stop alarm when power resumes. To overcome this problem, here is an automatic circuit that sounds an alarm for a few seconds only after power resumes. Author's prototype is shown in Fig. 1.

Circuit and working

Fig. 2 shows the circuit diagram of the automatic power-resumption alarm. It is built around quad NAND gate CMOS IC 4011 (IC1), timer NE555 (IC2) and transistor BC547 (T1). IC 4011 has four NAND gates but in this circuit we have used only two. A 9V power adaptor may be used for this circuit.

Whenever power failure occurs this circuit does not sound any alarm. But when power resumes it produces an alarm sound for about 11 seconds. This sound duration is determined by NE555 timer (IC2) configured in monostable mode. One-mega-ohm resistor (R4) and 10µF, 25V electrolytic capacitor (C2) are the timing compo-



Fig. 1: Author's prototype

PARTS LIST

Semiconductors:

- IC1 - 4011 CMOS quad NAND gate
- IC2 - NE555 timer
- T1 - BC547 npn transistor

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 1.8-kilo-ohm
- R2 - 10-ohm
- R3, R4 - 1-mega-ohm

Capacitors:

- C1 - 0.1µF ceramic disk
- C2 - 10µF, 25V electrolytic
- C3 - 0.01µF ceramic disk

Miscellaneous:

- CON1 - 2-pin terminal connector
- PZ1 - Piezo buzzer
- 9V power adaptor

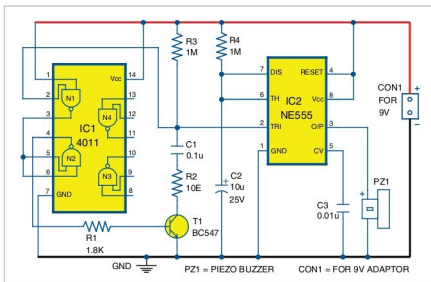


Fig. 2: Circuit diagram of the automatic power-resumption alarm

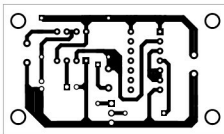


Fig. 3: Actual-size PCB pattern of the automatic power-resumption alarm

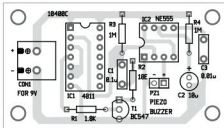


Fig. 4: Component layout of the PCB

nents. Duration of the alarm can be increased or decreased by changing the values of R4 and/or C2. This is determined by the relationship:

$$T = 1.1 \times R4 \times C2 \text{ in seconds}$$

Whenever power resumes, input pins 1 and 2 of NAND gate N1 become high, so NAND gate N2 output pin 4 also becomes high. This high output switches on transistor T1. IC2 receives a trigger pulse from T1 for a short time. Hence, the monostable circuit is switched on to produce an alarm.

Construction and testing

An actual-size, single-side PCB pattern for the automatic power-resumption alarm is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on the PCB, enclose it in a suitable box.

This is an easy-to-assemble circuit. Its alarm sound can be increased by using an additional amplifier circuit. **EFY**



N. Pugalum Perumal holds a diploma in electrical and electronics. He is a design enthusiast